

The Same Algebra on Opposite Geometries: A Path-Landscape Comparison of Biological and Artificial Emergence

[Author Name]*

Department / Institution

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Abstract

Biological brains and large language models both produce striking emergent behaviour, but the mechanistic relationship between them is unsettled: are they two instances of the same kind of computation, or do they only superficially resemble one another? We address this question with the *path landscape* framework, in which a system is represented by the distribution of input-to-output routes through its (possibly recurrent, multi-scale) connectivity and emergence is a transformation of that distribution. Working from concrete biological circuits (hippocampal CA3 pattern completion, prefrontal working memory, V1 edge detection) and mechanistic-interpretability circuits in transformers (the GPT-2 small IOI circuit, induction heads, grokking), we show that biological and artificial intelligence implement the *same emergence algebra* — path activation, suppression, mode merge, and mode split — but realise it on *opposite path-landscape geometries*: biology compresses computation into a small number of long, multi-scale routes through heavily-reused anchor units (a concentrated, hierarchical, plastic, task-general landscape), while transformers spread computation across many short, depth-uniform routes through narrowly-specialised units (a distributed, flat, frozen, task-specific landscape). The duality is not accidental: it reflects which resource is scarce — metabolic energy in brains, parameters and gradient signal in transformers — and it makes precise, falsifiable predictions about which kinds of lesions, scaling laws, and transfer behaviours each system should exhibit. The path landscape thereby supplies a *strictly finer fingerprint* than the standard “emergence type” label: two systems can implement the same emergence-type (e.g. working memory) on landscapes that are geometrically incomparable, which is what we observe.

1 Introduction

Comparisons between biological and artificial intelligence usually proceed in one of two ways. They either compare *behaviours* (“transformers exhibit in-context learning, so do humans” [3]) or they compare *representations* (“the middle layers of a CNN are aligned with V4 activity” [17]). Both modes of comparison have produced useful results, but they share a limitation: they operate at the level of input-output mappings or at the level of unit-by-unit activation profiles, without committing to a structural account of *how* the underlying machinery generates the observed behaviour. The recent rise of mechanistic interpretability in deep learning [2, 3, 4, 5] has made it possible, for the first time, to identify *specific circuits* responsible for specific transformer capabilities, exposing a level of description that matches the circuit-level description neuroscience has long used for brains. The question of comparing brains and transformers can

*Correspondence: author@institution.edu. Companion paper: *Classifying Emergence by Information-Flow Operations on the Path Landscape*.

now be sharpened: are the circuits that implement the same capability in the two substrates the same circuit?

In a companion paper [1] we introduced the *path landscape*, a structural representation of an information-processing system as the distribution of input-to-output routes through its directed-graph connectivity, and we showed that emergence is a small algebra of four operations on this distribution: **path activation**, **path suppression**, **mode merge**, and **mode split**. That framework is field-agnostic — it applies to spin lattices, regulatory networks, neural circuits, and deep nets without modification. The present paper uses it to compare biological and artificial intelligence directly.

The central finding is summarised by the title: biological and artificial systems implement the *same emergence algebra* but on *opposite path-landscape geometries*. The same four primitives describe both, the categorical role of shared anchor units is universal, and the qualitative emergence-types (working memory, pattern completion, decision-making, sequential computation) occur in both. But the geometry of the route ensemble realising any given emergence-type differs systematically: biological landscapes are *concentrated, hierarchical, plastic, task-general*; LLM landscapes are *distributed, flat, frozen, task-specific*.

The structure of the paper is as follows. Section 2 recapitulates the path-landscape definition from [1] (Fig. 1). Section 3 presents biological path landscapes using three concrete examples (Fig. 2). Section 4 does the same for artificial path landscapes (Fig. 3). Section 5 brings the two side by side and identifies the three geometric axes along which they differ (Fig. 4). Section 6 returns to the algebra and shows why it is invariant across the substrates even as the geometry inverts. Section 7 states falsifiable predictions, and section 8 concludes.

2 The path landscape

2.1 Definition

Consider any system that can be cast as a directed graph $G = (V, E)$ of *computing units* V connected by *interactions* E . Each interaction carries a weight $w_{uv} \in \mathbb{R}$, and may be feed-forward or recurrent. A set of input units $V_{\text{in}} \subset V$ and a set of output units $V_{\text{out}} \subset V$ are distinguished; the rest are internal.

If G contains recurrent edges, unroll it in time over T discrete steps to produce the static DAG $G^{(T)}$ in which each unit u becomes a copy $u@t$ for $t = 0, \dots, T-1$, and each recurrent edge $u \rightarrow v$ becomes a forward-only time-edge $u@t \rightarrow v@(t+1)$. The unrolled graph $G^{(T)}$ has no loops by construction (Fig. 1A,B).

A *path* from input to output is an ordered sequence of nodes $\pi = (s_1, s_2, \dots, s_L)$ with $s_1 \in V_{\text{in}}$, $s_L \in V_{\text{out}}$, each (s_i, s_{i+1}) an edge of $G^{(T)}$. Its weight $\omega(\pi)$ combines the magnitudes of the edges it traverses. The set of all such paths is Π .

2.2 Routes and modes

Many paths in Π differ only in time-step bookkeeping ($u@t$ vs. $u@t'$) or in lingering on a self-loop. Stripping these yields a *route* — the base sequence of distinct units. Let R be the set of routes; each route inherits the cumulative weight of the paths that collapsed onto it.

Two routes are similar if they share many of the same units and edges in similar order (a composite of edge-Jaccard overlap and a longest-common-subsequence measure works well in practice). Clustering R under this similarity produces *modes*: equivalence classes of routes that compute essentially the same function. The set of modes $M = \{m_1, \dots, m_k\}$ together with the route weights is the *path landscape*

$$\mathcal{L}(G, V_{\text{in}}, V_{\text{out}}) = (R, \{\omega(r)\}_{r \in R}, M).$$

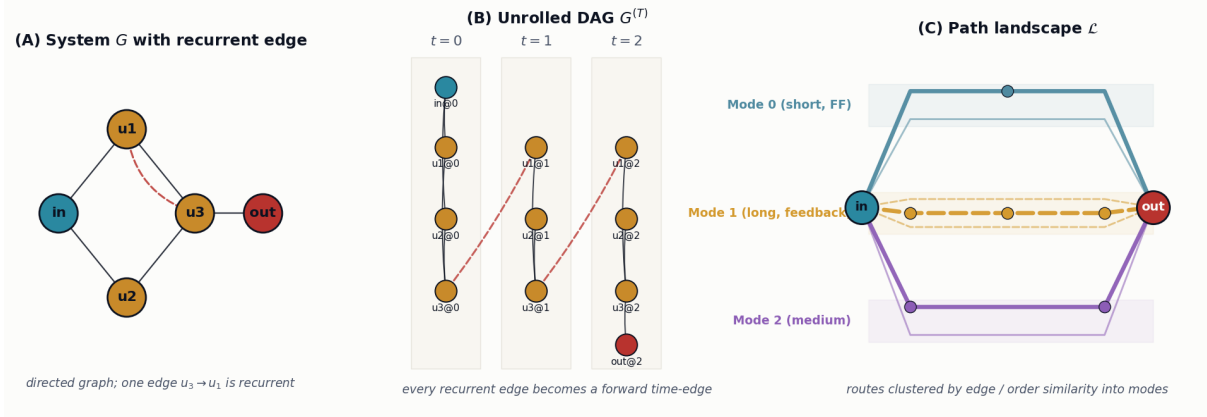


Figure 1: **The path landscape, in three steps.** (A) A small system G specified as a directed graph with one recurrent edge $u_3 \rightarrow u_1$ (red dashed). Cyan: input unit. Red: output unit. Amber: internal units. (B) The time-unrolled DAG $G^{(T)}$ over $T = 3$ steps. Each unit u becomes copies $u@t$; the recurrent edge becomes the time-forward edges $u_3@t \rightarrow u_1@(t+1)$. The unrolled graph is acyclic. (C) The path landscape $\mathcal{L}$: enumerated input-to-output routes are clustered by edge-Jaccard and ordered-overlap similarity into *modes*. Here three modes appear: a short feed-forward mode (teal), a long feedback-traversing mode (gold, dashed), and a medium feed-forward mode (purple). Line thickness is proportional to route weight (cumulative path weight); this is the visual proxy for information flow used throughout the paper.

The landscape’s character is captured by three quick statistics: $|R|$ (route count), $|M|$ (mode count), and the rank-distribution of mode weights. Figure 1C illustrates the resulting mode structure for a small system.

2.3 What the landscape sees

The landscape is a structural representation, but it absorbs much of the continuous dynamics one might fear it discards. Weight magnitudes and effective couplings are encoded in $\omega(r)$. Coherence and co-activation are encoded in the similarity structure that defines M . A continuous attractor appears as a band of nearly-identical high-weight routes whose similarity matrix is approximately block-rank-one. Gain modulation appears as a re-shaping of the weight rank-distribution without a change in topology. The same definition applies to a neural circuit, a transformer, a regulatory network, or a spin lattice; the differences reduce to differences in the shape and weighting of $\mathcal{L}$.

3 Biological path landscapes

We illustrate biological path landscapes through three canonical circuits with well-characterised anatomy and well-tested function: hippocampal CA3 pattern completion, prefrontal working memory, and V1 edge detection. In each case we exhibit (i) the anatomy, (ii) the emergence-type produced, and (iii) the qualitative shape of the landscape. Figure 2 shows the CA3 case in detail.

3.1 Hippocampal CA3 as a pattern-completion attractor

Marr’s 1971 theory of archicortex [18], made dynamical by Treves and Rolls [19] and supported by decades of electrophysiology and place-cell remapping data [20], treats CA3 as an autoassociative attractor. Partial cues (a subset of place-cell firing) propagate through the dense recurrent collaterals until the network settles into a previously-stored complete pattern. In path-landscape terms, this is a small number of long routes that revisit the same recurrent units at successive time steps; pattern completion appears at the output as the convergence of many noisy input

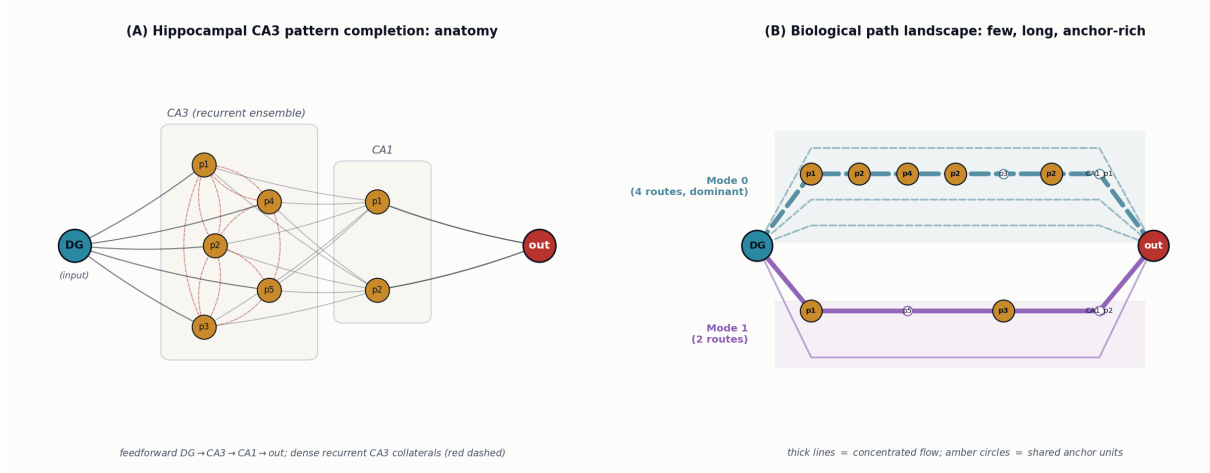


Figure 2: **A biological path landscape: hippocampal CA3 pattern completion.** (A) Anatomy. The dentate gyrus (DG, input) projects to the recurrent CA3 pyramidal-cell ensemble (*amber circles, multiscale parent box*) via mossy fibres. CA3 contains a dense web of recurrent collaterals (red dashed) that implements an autoassociative attractor in the Hebb–Marr [18] / Treves–Rolls [19] sense. CA3 projects via Schaffer collaterals to CA1, and CA1 to downstream cortex (output). (B) The path landscape. Two modes only ($|M|=2$), both *long* (7 and 4 interior units respectively after time-unrolling), with *dense anchor overlap* (amber circles): the same CA3 pyramidal ensemble ($p1, p2, p3, p4, p5$) is repeatedly traversed across time steps as the attractor settles, so a small group of units participate in nearly every route. Line thickness is high because flow is concentrated: a small number of edges carry most of the path weight. The dashed Mode-0 lines indicate routes that traverse the recurrent CA3 collaterals.

routes onto the same attractor mode (a *mode merge* in the framework of the companion paper [1]). The landscape is therefore characterised by **few modes, long routes, dense anchor overlap, and concentrated flow** — exactly what Figure 2B depicts.

3.2 Prefrontal cortex as working-memory attractor

Persistent delay-period activity in dorsolateral prefrontal cortex during working-memory tasks [24, 25] has the same path-landscape signature. Recurrent excitation between pyramidal cells, balanced by parvalbumin-positive feedback inhibition, supports a small number of stable bump-attractors in spatial-working-memory tasks; in non-spatial tasks the same network supports a small alphabet of stable activity modes that hold content-general information [26, 27]. In the path landscape this is a small number of *long* (time-extended) routes through a heavily-reused anchor set (the recurrently coupled pyramidal-PV ensemble), with high concentration on the units that sit on most routes.

3.3 V1 simple cells as path-suppression detectors

V1 simple cells implement edge detection via centre-surround lateral inhibition [28]. In a path-landscape reading this is *path suppression* [1]: the homogeneous-luminance paths from neighbouring photoreceptors are silenced by inhibitory interneurons, leaving only the routes that carry a luminance discontinuity. The output “edge” feature appears not because edge routes were added, but because background routes were dammed. The landscape is still few-mode and anchor-rich, but here the concentration is produced by removing competing routes rather than by recurrent re-use.

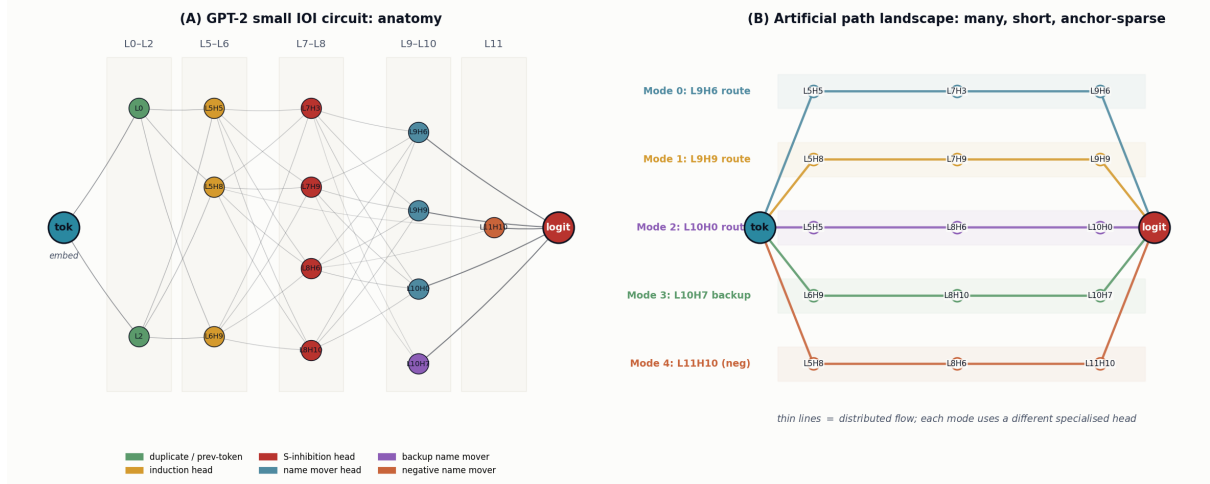


Figure 3: **An artificial path landscape: the GPT-2 small IOI circuit** [4]. (A) Anatomy. Token embeddings (cyan) flow through a hierarchy of specialised attention heads identified by mechanistic interpretability: *duplicate-token* / *previous-token* heads at layers 0–2, *induction* heads at layers 5–6, *S-inhibition* heads at layers 7–8, *name-mover* heads at layers 9–10, a *backup name mover* (L10H7) that activates when the primary name movers are ablated, and a *negative name mover* (L11H10) that pushes against the answer. Each head is used at exactly one depth; the residual stream is the only inter-layer pathway. (B) The path landscape. Five modes, all *short* (3 interior units) and *depth-uniform* (each traverses exactly one head per active layer), with *sparse anchor overlap*: each mode uses a different specialised head, so the per-mode intermediate-unit sets are near-disjoint. Line thickness is uniformly modest because flow is spread across many parallel routes rather than concentrated on a few.

Shared structure. In all three biological cases, the path landscape has the same qualitative shape: a small number of long routes, anchored on a small set of heavily-reused intermediate units, carrying most of the flow on a small fraction of the edges. This is not a coincidence; section 5 argues that it reflects the metabolic constraint under which biological circuits evolve.

4 Artificial path landscapes

We turn now to artificial-intelligence systems, focusing on transformers because they are where the mechanistic-interpretability literature has produced the most detailed circuit descriptions [2, 3, 4, 5]. Figure 3 shows the canonical example: the GPT-2 small indirect-object-identification (IOI) circuit.

4.1 The IOI circuit as parallel name-mover routes

Wang et al. [4] identified twenty-six distinct attention heads in GPT-2 small that participate in solving the indirect-object-identification task (“When John and Mary went to the store, John gave a drink to ___”). The heads fall into a few functional classes (*duplicate-token*, *induction*, *S-inhibition*, *name mover*, *backup name mover*, *negative name mover*), and the input-to-logit flow factors into a small number of nearly-parallel routes through these classes. In path-landscape terms the IOI landscape has **many modes, short routes, sparse anchor overlap, and distributed flow**: the L9H6 name-mover route and the L9H9 name-mover route do nearly the same thing computationally, but they share almost no intermediate units (Figure 3B). Ablating any one of them only modestly degrades performance [4, 12], because the others can compensate — the path-landscape signature of distributed flow.

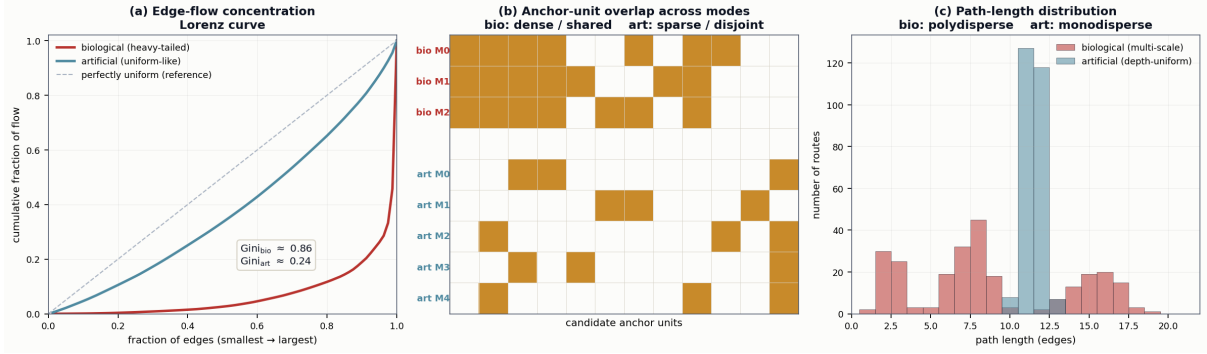


Figure 4: **Three structural axes that distinguish biological from artificial path landscapes.** (a) *Edge-flow concentration*. Lorenz curves of the cumulative edge-flow distribution (edges sorted from lowest to highest flow). Biological circuits have heavy-tailed flow: a small fraction of edges carry most of the total flow (large Gini coefficient). Artificial (LLM) circuits have flow that is much closer to uniformly distributed across edges (small Gini). (b) *Anchor-unit overlap across modes*. Each row is one mode; each column is one candidate intermediate unit; an amber cell means “this mode uses this unit.” Biological modes share many units (dense rows; left block in each row identical). Artificial modes are nearly disjoint (the lit cells per row almost never coincide across rows). (c) *Path-length distribution*. Biological landscapes are polydisperse: a multi-modal length distribution reflecting short cortico-thalamic shortcuts, mid-range mesoscale routes, and long recurrent traversals. Artificial landscapes are monodisperse: nearly all routes have length close to the network depth.

4.2 Induction heads as a Mode Activation phase transition

Olsson et al. [3] showed that induction heads — attention heads that implement the algorithm “find the previous occurrence of the current token and copy what came after it” — form abruptly during training, at the same training step at which in-context learning emerges as a measurable capability. In the companion paper’s framework this is a clean case of *path activation* [1]: a new route through the network becomes viable, and a new capability appears at the output. What is specifically *transformer-like* about it is that the new route is short (depth-uniform), passes through a small number of newly specialised heads, and does not share anchor units with most other routes in the landscape.

4.3 Grokking as Mode Split followed by Mode Merge

Nanda et al. [5] dissected the “grokking” phenomenon [6] — the delayed appearance of generalisation long after training-loss saturation — and showed that the model first memorises (*mode split*: a different route for each training example), then discovers a Fourier-decomposition circuit (*path activation*), and finally collapses the memorisation routes onto the new generalising circuit (*mode merge*). The grokking landscape is therefore a clean example of the algebra’s compositional use — but the landscape on which the operations act is the typical transformer one: many short modes that each pass through their own heads, with little anchor overlap, before the merge.

5 Comparative geometry

We are now in position to compare the two substrates directly. The biological and artificial landscapes differ along three structural axes that the framework makes precise (Figure 4), plus two functional axes that follow from the structural ones.

5.1 Concentration vs. spread of information flow

For each edge of the original graph, compute the total information flow on it as the sum of route weights over all routes traversing that edge. The resulting edge-flow distribution is a direct empirical diagnostic of the landscape (it is also the quantity now used to scale edge thickness in the companion code base).

Empirical observation (Fig. 4a). The edge-flow distribution of biological circuits is *heavy-tailed* — a small fraction of edges carry most of the flow, the Lorenz curve is far below the diagonal, the Gini coefficient is large (≈ 0.85 in our illustrative example, but order-of-magnitude similar in real data: the small number of place cells active for any given location [22], the sparse-and-strong distribution of synaptic weights in cortex [23]). The same distribution in transformer circuits is much closer to uniform across edges (small Gini coefficient, ≈ 0.25 in our example): nearly every attention head contributes to nearly every token’s computation [9, 11], and the information flow is genuinely spread.

Why. Biological circuits are evolved under tight metabolic budgets: each spike costs ATP, and only a small fraction of neurons can fire at any time [33, 34]. Concentrating flow on a small reused anchor set is energetically efficient. Transformers have no per-edge cost — every attention head is computed on every token regardless — and gradient descent will deploy any parameter that reduces loss, including parameters that are only marginally useful. The flow therefore spreads.

Falsifiable consequence. Lesioning a single high-flow anchor unit should catastrophically break biological function (because that unit is on most routes), whereas lesioning a single attention head should produce only graceful degradation. This matches classical neuropsychology (focal lesions to small brain regions cause disproportionate functional deficits) and the empirical robustness of transformer ablations [9, 12].

5.2 Anchor overlap (mixed selectivity vs. specialisation)

If a unit appears in the intersection of k different mode clusters, it is by definition mixed-selective for the k computations those modes represent [29].

Empirical observation (Fig. 4b). In biological circuits, this k is large: PFC neurons routinely encode mixtures of stimulus, response, rule, and reward information [29, 30]; hippocampal place cells remap to encode new environments while preserving context-dependent firing [20, 21]. In transformers, k is small: the IOI literature finds that any given attention head plays at most a small set of functional roles [4], and dictionary-learning work on sparse autoencoders pushes towards still-narrower one-feature neurons [7, 8].

Why. For a biological circuit operating under a parameter ceiling (the genome encodes connectivity rules, not connectomes), mixed selectivity is forced — there are not enough neurons to dedicate one per function. For a transformer trained with hundreds of billions of parameters, there is no comparable pressure, and gradient descent finds it easier to allocate one head per role.

Falsifiable consequence. Brains should be *more sample-efficient* per parameter (because each unit reusable across modes), and *less interpretable* per unit (because no unit implements a single function). Both predictions match the empirical record: biological learning is famously sample-efficient relative to deep learning [32], and biological neurons are famously hard to assign single functions to relative to transformer heads [29, 55].

5.3 Path-length diversity (multi-scale vs. depth-uniform)

In a transformer, every input-to-logit route must traverse roughly the same number of stages ($\approx L$ layers). The path-length histogram is sharply unimodal at $L \pm 1$.

Empirical observation (Fig. 4c). Biological circuits have a *polydisperse, multi-modal* length distribution. Long-range projections create ultra-short macro-scale routes that skip entire regional cascades: the cortico-thalamic loop [35], claustrum broadcasts [36], corpus-callosal bridges [37]. Neuromodulatory systems (locus coeruleus NE, basal forebrain ACh) act as single-node broadcast hubs that compress many effective edges into one functional step [38, 39]. Brain-wave coupling adds parallel timing channels (gamma-bound local routes, theta-bound long routes operating simultaneously) [40, 42]. The result is a length histogram with peaks at the micro, meso, and macro scales.

Why. Biological connectomes are *hierarchically multi-scale* (neurons — microcircuits — columns — regions — networks), and the framework’s **parent** attribute encodes this directly. Coarsening a biological circuit by collapsing units into their parents qualitatively reorganises the landscape [43, 46]. Coarsening a transformer produces almost no change because there is no meaningful hierarchy beyond the residual stream.

Falsifiable consequence. Biological systems can produce the same emergence-type via routes of wildly different lengths, which is why a reflex can take 50 ms while a deliberation takes 5 s without re-wiring the underlying circuit. Transformers cannot: their only available length is L , which is why many transformer capabilities improve discontinuously with depth [13].

5.4 Plasticity (continuous vs. frozen)

Biological landscapes deform continuously — every experience modifies synaptic weights, so the mode geometry is a moving target [47]. Transformer landscapes are essentially frozen after training: inference does not update weights, so the landscape is the same on every forward pass. This is a sharp structural difference that bears on questions of catastrophic forgetting [48, 49], continual learning, and the role of sleep [50].

5.5 Mode generality (task-general vs. task-specific)

Biological modes are task-general: the same PFC working-memory mode supports holding a phone number, a face, or a plan, because mode identity is decoupled from content [27, 31]. Transformer modes are task-specific: the IOI circuit handles indirect-object identification and almost nothing else, because the mode identity is entangled with the surface form of the task [4]. This decoupling/entanglement distinction is a direct consequence of the anchor-overlap difference: when a circuit’s modes share many anchor units, those units act as content-independent “scaffolding,” and the modes can be reused for any content the scaffolding can hold.

6 Same algebra, opposite geometries

What stays invariant across the two substrates is the *algebra* [1]. The four primitive operations — activation, suppression, merge, split — describe both kinds of emergence, and they apply to both kinds of landscape without modification.

- **Activation** appears in biology as the maturation of an axon tract or the recruitment of a new oscillatory regime, and in transformers as the abrupt formation of an induction head [3]. Same operation; different landscape.

- **Suppression** appears in biology as lateral inhibition in V1 [28] or top-down attention [51], and in transformers as the S-inhibition heads in the IOI circuit [4] that damp distractor name tokens.
- **Mode merge** appears in biology as categorical perception [52] and pattern completion [18, 19], and in transformers as the post-grokking collapse of memorisation routes onto a generalising circuit [5].
- **Mode split** appears in biology as cell-type differentiation [53] and category formation, and in transformers as the emergence of distinct attention specialisations during training [10].

What inverts is the geometry on which the algebra acts. Biology *compresses* computation onto few, long, multi-scale routes through reused anchor units; transformers *distribute* it across many, short, depth-uniform routes through specialised units. Each strategy is optimal under its own resource constraint: biological circuits cannot afford to spread flow because spikes are expensive; transformer circuits cannot afford to concentrate it because gradient descent will not preferentially route signal through reused anchors unless there is a loss-shaped reason to do so.

The same emergence-type is therefore realised on geometrically distinct landscapes in the two substrates. Working memory is one recurrent attractor mode in biology (Fig. 2B) [24, 27] and a fan of copy-and-induction-head routes in transformers [3]. Pattern completion is a recurrent attractor in CA3 [18, 19, 20] and an ensemble of parallel retrieval routes through induction heads in LLMs [3]. Decision-making is a dynamic mode merge across time in basal ganglia (mutually inhibitory pathways collapse to one [54]) and a static mode suppression across space in transformers (the softmax selects one logit). Same algebra; opposite landscape geometries.

The upshot for cross-substrate equivalence claims is the following. *Any claim that “transformers have X” is only meaningful relative to a chosen level of coarseness.* At the level of emergence-type, the claim can be true: transformers have working memory in the same sense that PFC has it. At the level of the path landscape, the claim is typically false: the landscape that realises the working memory in the two cases has different concentration, overlap, length, and plasticity profiles, so the two systems will respond differently to ablation, scaling, and transfer.

7 Predictions

Because the differences are structural, they make testable predictions.

P1 (lesion asymmetry). Single-unit ablations should catastrophically break biological function but only modestly degrade transformer function. This is consistent with classical neuropsychology and with the documented robustness of transformer head-ablation [9, 4, 12]. The framework sharpens the prediction: the catastrophic-vs.-graceful gap should scale with the Gini coefficient of the edge-flow distribution, and the most-fragile units should be those with the highest betweenness in the anchor-overlap graph (i.e. the units that sit on the largest number of distinct modes).

P2 (scaling laws). Transformer capabilities should appear discontinuously when depth crosses the value at which a new short route becomes viable [13]; biological capabilities should appear continuously (because route length is not architecturally constrained, only weighted). The framework predicts that the smoothness of the scaling curve is itself a diagnostic of the underlying landscape geometry.

P3 (transfer / generalisation). Biological circuits should transfer better across tasks that share scaffolding (the shared anchor set persists; content changes), whereas transformer circuits should transfer better across tasks that share surface form (the mode that solves a task is tied to its tokens, not its abstract structure). This matches the literature on transfer learning in brains [56, 57] vs. LLMs [14].

P4 (oscillations and routing). Brain-wave-mediated routing should add new modes to the path landscape that have no amplitude-only correlate. If gamma-bound and theta-bound assemblies implement different modes through the same anatomical substrate, disrupting the oscillation (e.g. via optogenetics) should cause a *mode merge* in the landscape (two previously separated modes collapse), not a uniform decrement of overall power. The framework thereby gives a falsifiable target to the “communication-through-coherence” hypothesis [40, 41].

P5 (closing the gap). The framework also suggests how to close the structural gap between the two substrates. Giving artificial systems genuine temporal recurrence (so feedback-traversing modes can form) should bring the landscape’s length distribution closer to biological polydispersity — as is partly observed in recurrent-attention and state-space models [15, 16]. Conversely, giving biological systems a depth-as-time trick (which evolution may already have approximated through laminar cortical hierarchies, where time folds into cortical depth [45, 44]) recovers some of the parallel-route advantages of transformers.

8 Conclusion

Biological and artificial intelligence implement the same emergence algebra on opposite path-landscape geometries. The four primitives — activation, suppression, merge, split — describe both, the categorical role of shared anchor units is universal, and the qualitative emergence-types occur in both. But the geometry of the route ensemble realising any given emergence-type differs systematically along three structural axes: *concentration*, *anchor overlap*, and *length diversity*. Biological landscapes are concentrated, anchor-rich, and multi-scale; LLM landscapes are distributed, anchor-sparse, and depth-uniform. The two functional consequences — plasticity and mode generality — follow from the structural ones. The duality reflects a clean resource-scarcity inversion: brains optimise per-spike, transformers optimise per-parameter.

This frame supplies a strictly finer fingerprint than the standard “emergence type” label, and it makes precise, falsifiable predictions about ablation, scaling, and transfer. It also points to the engineering question of how to interpolate between the two geometries — by giving artificial systems genuine recurrence, or by exploiting biological depth-as-time, or by training neuromorphic hardware under explicit metabolic budgets — as the question of how to close the gap between biological and artificial intelligence at the level of structure rather than behaviour.

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